

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Display
Manufacturer: Sharp
Firmware Version: N/A
Model(s): PN-LE601, PN-LE701, PN-LE801, PN-LE901

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.01.0000-b010	2.1.0

Version History

Module Version	Date	Notes
1_0_5_2	1/28/2019	Fixed Update Nonetype error.
1_0_5_1	10/17/2018	Fixed Power Off command
1_0_5_0	8/3/2018	Updated several commands and fixed default Ethernet port
1_0_1_0	10/26/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: InterfaceName.Unidirectional = 'True'
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: InterfaceName.connectionCounter = 5

Supported Classes and Examples

SerialClass
InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='PN-LE901')
SerialOverEthernetClass
InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='PN-LE901')
EthernetClass
InterfaceName = ModuleName.EthernetClass('192.168.254.254', 10002, Model='PN-LE901')

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio	Value 'Side Bar (AV)' 'Stretch (AV)' 'Stretch (PC)' 'Original'	Value 'S. Stretch (AV)' 'Normal (PC)' 'Full Screen (AV)'	Value 'Zoom (AV)' 'Dot by Dot (PC)' 'Auto'
# AspectRatio example InterfaceName.Set('AspectRatio', 'Side Bar (AV)')			
Command ATVChannelDirectCommand	Value '1' - '135'		
# ATVChannelDirectCommand example InterfaceName.Set('ATVChannelDirectCommand', None)			
Command AVMode	Value 'Standard' 'Dynamic (Fixed)'	Value 'Movie' 'Dynamic'	Value 'User' 'PC'
# AVMode example InterfaceName.Set('AVMode', 'Standard')			
Command ChannelStep ¹	Value 'Up'	Value 'Down'	
# ChannelStep example InterfaceName.Set('ChannelStep', 'Up')			
Command ClosedCaption	Value None		
# ClosedCaption example InterfaceName.Set('ClosedCaption', None)			
Command DTVChannelAirCommand	Value '100' - '9999'		
# DTVChannelAirCommand example InterfaceName.Set('DTVChannelAirCommand', None)			
Command DTVChannelCable1Command	Value '1.0' - '999.999'		
# DTVChannelCable1Command example InterfaceName.Set('DTVChannelCable1Command', None)			

Command DTVChannelCable2Command and	Value '0' - '16383'		
# DTVChannelCable2Command example InterfaceName.Set('DTVChannelCable2Command', None)			
Command Input	Value 'HDMI 1' 'Video' 'TV'	Value 'HDMI 2' 'Component'	Value 'HDMI 3' 'PC'
# Input example InterfaceName.Set('Input', 'HDMI 1')			
Command MenuNavigation	Value 'Menu' 'Exit' 'Left'	Value 'Enter' 'Up' 'Right'	Value 'Return' 'Down'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Menu')			
Command Mute	Value 'On'	Value 'Off'	
# Mute example InterfaceName.Set('Mute', 'On')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command Volume	Value 0 to 100 in steps of 1		
# Volume example InterfaceName.Set('Volume', 100)			

¹ This command will switch the Input to TV

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio	Value 'Side Bar (AV)' 'Stretch (AV)' 'Stretch (PC)' 'Original'	Value 'S. Stretch (AV)' 'Normal (PC)' 'Full Screen (AV)'	Value 'Zoom (AV)' 'Dot by Dot (PC)' 'Auto'
# AspectRatio examples InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command AVMode	Value 'Standard' 'Dynamic (Fixed)'	Value 'Movie' 'Dynamic'	Value 'User' 'PC'
# AVMode examples InterfaceName.Update('AVMode') Value = InterfaceName.ReadStatus('AVMode') InterfaceName.SubscribeStatus('AVMode', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command Input	Value 'HDMI 1' 'Video'	Value 'HDMI 2' 'Component'	Value 'HDMI 3' 'PC'
# Input examples InterfaceName.Update('Input')			

Global Scripter Module Communication Sheet

<pre>Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)</pre>		
Command Mute	Value 'On'	Value 'Off'
<pre># Mute examples InterfaceName.Update('Mute') Value = InterfaceName.ReadStatus('Mute') InterfaceName.SubscribeStatus('Mute', None, FeedbackHandler)</pre>		
Command Power	Value 'On'	Value 'Off'
<pre># Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>		
Command Volume	Value 0 to 100 in steps of 1	
<pre># Volume examples InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)</pre>		

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 9600

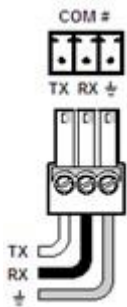
Data Bits: 8

Parity: None

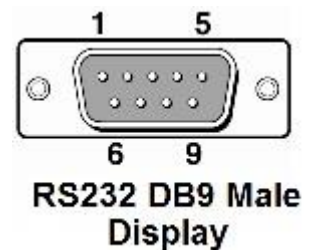
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		3	TxD
RxD		2	RxD
GND		5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	10002
Logon Credentials Supported:	No
Multi-Connection	Undetermined
Capabilities:	
Port Changeability:	Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Aspect Ratio Auto	WIDE10 \x0D
Aspect Ratio Dot by Dot (PC)	WIDE8 \x0D
Aspect Ratio Full Screen (AV)	WIDE9 \x0D
Aspect Ratio Normal (PC)	WIDE5 \x0D
Aspect Ratio Original	WIDE11 \x0D
Aspect Ratio S. Stretch (AV)	WIDE2 \x0D
Aspect Ratio Side Bar (AV)	WIDE1 \x0D
Aspect Ratio Stretch (AV)	WIDE4 \x0D
Aspect Ratio Stretch (PC)	WIDE7 \x0D
Aspect Ratio Zoom (AV)	WIDE3 \x0D
ATV Channel Direct Command None	DCCH001 \x0D
AV Mode Dynamic	AVMD6 \x0D
AV Mode Dynamic (Fixed)	AVMD5 \x0D
AV Mode Movie	AVMD2 \x0D
AV Mode PC	AVMD7 \x0D
AV Mode Standard	AVMD1 \x0D
AV Mode User	AVMD4 \x0D
Channel Step Down	CHDW1 \x0D
Channel Step Up	CHUP1 \x0D
Closed Caption None	CLCP1 \x0D
DTV Channel Air Command None	DA2P0100\x0D
DTV Channel Cable 1 Command None	DC2U001 \x0DDC2L001 \x0D
DTV Channel Cable 2 Command None	DC100000\x0D
Input Component	IAVD5 \x0D
Input HDMI 1	IAVD1 \x0D
Input HDMI 2	IAVD2 \x0D
Input HDMI 3	IAVD3 \x0D
Input PC	IAVD6 \x0D
Input TV	ITVD0 \x0D
Input Video	IAVD4 \x0D
Menu Navigation Down	RCKY42 \x0D
Menu Navigation Enter	RCKY40 \x0D
Menu Navigation Exit	RCKY46 \x0D
Menu Navigation Left	RCKY43 \x0D
Menu Navigation Menu	RCKY38 \x0D
Menu Navigation Return	RCKY45 \x0D

Global Scripter Module Communication Sheet

Revision: 8/3/2018

Menu Navigation Right	RCKY44 \x0D
Menu Navigation Up	RCKY41 \x0D
Mute Off	MUTE2 \x0D
Mute On	MUTE1 \x0D
Power Off	POWR0 \x0D
Power On	POWR1 \x0D
Volume 0	VOLM000 \x0D
Volume 100	VOLM100 \x0D

Appendix B. Update Commands

Aspect Ratio	WIDE????\x0D
AV Mode	AVMD????\x0D
Input	IAVD????\x0D
Mute	MUTE????\x0D
Power	POWR????\x0D
Volume	VOLM????\x0D